

Auditing Political Exposure Bias in News Recommendation

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This project investigates whether a news recommender can lead users to become surrounded by what they interact with, especially in a political ideology setting. Using interaction logs and an ideology labeling pipeline, the report analyzes the distribution of political and ideological content in the candidate slate versus the ranked Top-K recommendations, and runs short-horizon event-style analyses to test whether exposure shifts after specific types of clicks. The results indicate systematic exposure imbalances and suggest that ranking can amplify differences between what is available and what is shown. Finally, simple, supply-aware mitigation strategies are evaluated, and additional longer-horizon interventions and potential social impacts are discussed.

I. INTRODUCTION AND MOTIVATION

News recommenders operate at the intersection of machine learning and democratic life: what users see can influence knowledge, opinions, and perceived social consensus. While personalization improves relevance, it can also create uneven exposure patterns. Two socio-technical risks motivate this study:

- **Exposure imbalance across users:** Politically engaged users can receive disproportionately more political content, while less engaged users may be under-exposed, reducing their chance to encounter civic information.
- **Under-representation of minority viewpoints:** If one ideology class is scarce or systematically ranked lower, Top- K feeds may miss that viewpoint, creating a skewed political spectrum.

This project addresses the question: How do ranking and data limitations interact to produce political exposure bias, and how far can we mitigate it using cheap post-processing re-ranking without harming recommendation utility?

II. BACKGROUND AND RELATED WORK

A. Methods, techniques and tools used in this work

This project follows an *audit-then-mitigate* methodology applied to news recommendation logs.

First, I profile the interaction data and build user cohorts.

Second, I attach ideology labels to political items using an ideology mapping/prediction pipeline and compute exposure metrics at two stages: (i) the *candidate slate* and (ii) the ranked *Top-K*.

Third, I run a short-horizon event-style analysis (from impression t to $t+1$) to test whether exposure composition changes after different click types. Finally, I evaluate post-processing reranking mitigations and report both exposure effects and ranking-utility metrics (MRR@K

and nDCG@K)¹. All experiments are implemented in Python using Jupyter notebooks with NumPy/Pandas.

B. Related work

Fairness-of-exposure research formulates constraints and objectives directly over ranked lists². In parallel, work on popularity bias and feedback loops shows how engagement-based recommendation can amplify dominant content over time³. Prior literature also emphasizes the distinction between *supply constraints* and *ranking effects* when evaluating fairness in recommender systems. Compared to these approaches, this project contributes an empirical audit separating supply vs. ranking effects and a lightweight mitigation study using post-processing reranking.

III. METHODOLOGY AND EXPERIMENTAL SETUP

A. Overview of the audit pipeline

This project implements a notebook-driven auditing pipeline to study whether a news recommender can (i) create exposure imbalance across users and (ii) under-represent certain political ideologies (ex. left/center/right), potentially surrounding users with what they interact with.

B. Dataset and unit of analysis

I use the Microsoft News Dataset (MIND)⁴ with train/dev splits of news items and user interaction logs. The unit of analysis is an **impression**: a user is shown a slate of candidate news items. The audit extracts three impression-level objects: **candidates(*imp*)** (list of candidate news IDs shown/available for impression *imp*), **clicked(*imp*)** (set of clicked news IDs for impression *imp*) and **user(*imp*)** (the user ID associated with the impression).

C. Preprocessing and derived tables (Notebooks 01–02)

Notebook 01 (Sanity check) validates dataset integrity and confirms that IDs, impressions, and candidate slates can be joined consistently.

Notebook 02 (Click table construction) standardizes the logs into analysis-ready tables. In other words: this stage builds the *ground truth interface* for the remainder of the pipeline (who saw what, and what they clicked/interacted with).

You can see more details in the Appendix B.

D. User grouping and engagement signals (Notebook 03)

After the logs are standardized, **Notebook 03** constructs descriptive profiling statistics and defines cohorts to test whether exposure differs by user behavior. Users are partitioned into:

- **engaged**: users with sustained political interaction patterns (e.g., political clicks ≥ 3),
- **other**: all remaining users.

This grouping is necessary because the main fairness question is comparative: even if overall political exposure is low, the system may still allocate political content unevenly across cohorts.

E. Ranking baselines and offline utility evaluation (Notebook 04)

With cohorts defined, **Notebook 04** establishes baseline recommenders and evaluation metrics, so later mitigation changes can be interpreted as trade-offs rather than isolated numbers. Two baseline scoring functions are used throughout:

- **Popularity-based scoring**: ranks candidates using a popularity proxy derived from interactions.
- **Category-preference personalization**: ranks candidates using user-category affinity signals.

The metrics used here are available in the Appendix A 2 and are later reused to check whether exposure-mitigation reranking produces only small utility changes.

F. Temporal exposure diagnostics (Notebook 05)

After establishing cohorts and baseline rankers, **Notebook 05** investigates whether political exposure is purely a static property of the dataset or whether it behaves

like a *feedback process* over time. Concretely, I analyze exposure and interaction summaries across impressions in chronological order, focusing on (i) how **political exposure differs between engaged and other users**, (ii) whether **exposure remains stable across time** or appears to **concentrate for certain users** and (iii) whether **engagement correlates** with being shown **more political content**.

This notebook motivates later mitigations that are *history-aware* (e.g., balancing exposure across windows) rather than reacting to a single click.

G. Ideology labeling and label-quality audit (Notebook 06)

To analyze political exposure beyond a binary political/non-political split, I extend the pipeline with an ideology labeling component. This step assigns political items to ideology categories and provides the labels used in later notebooks.

a. Label source and outputs. I use an existing transformer-based classifier, `bucketresearch/politicalBiasBERT` (Hugging Face)⁵, to obtain 3-class ideology predictions {left, center, right} from the available news text (e.g., title and/or abstract). Items for which the model does not return a usable prediction are treated as *unlabeled*.

b. Coverage and imbalance reporting. Because automated ideology labels may be incomplete and the labeled subset may be skewed across classes, I explicitly compute and report **label coverage** and **class distribution** explained in Appendix C. These quantities are reported alongside downstream results so that any ideology-based analysis is interpreted with awareness of potential miss- ingness and class imbalance.

c. Manual audit. To validate the usefulness of the model-generated labels, I perform a small manual audit on a sample of labeled items, comparing the predicted ideology against a human judgment. I report basic agreement statistics (e.g., accuracy) to contextualize the reliability of the labels used in subsequent analyses.

H. Supply vs. ranking effects on ideology exposure (Notebook 07)

With label coverage understood, **Notebook 07** addresses the core question of *where* under representation emerges:

Does ideology imbalance come mainly from what is available (candidate slate), or from how the system ranks (Top-K)?

The methodology compares two distributions (Candidate and Top-K) computed over the same impressions. These are better explained in the Appendix D.

I. Short-horizon event study: $t \rightarrow t + 1$ (Notebook 07)

To test *immediate reinforcement*, I run an event-style analysis at the impression level:

- define an “event” at time t based on what ideology type was clicked,
- then compare the **candidate slate composition** at t versus the **next impression** $t + 1$ for the same user.

I split events into three cohorts (center, non-center and non-political clicks), and for each one I compute candidate-slate shares at t and at $t + 1$. More details on this topic in Appendix E.

J. Notebook 08 - Mitigation experiments and wrap-up

Goal: test simple, supply-aware mitigations and quantify trade-offs between exposure fairness and ranking utility.

I tested **exposure-gap mitigation** and **center-visibility mitigation**, described in Appendix F.

Outcome: the tested mitigations can substantially reduce measured exposure imbalance while causing negligible changes in offline utility metrics in this dataset, but their effectiveness is constrained by supply (center rarity/coverage).

IV. RESULTS AND DISCUSSION

This section presents the main empirical findings in the same logical sequence as the notebook pipeline. Rather than treating each notebook as an isolated experiment, I use the results of each stage to motivate the next stage: (i) build reliable impression-level structures, (ii) define user cohorts, (iii) establish baseline exposure and utility, (iv) validate ideology labels and quantify supply limitations, (v) separate *supply vs ranking* effects, (vi) probe short-horizon reinforcement, and (vii) evaluate mitigations under realistic constraints.

A. From raw logs to impression-level analysis objects (Notebooks 01 - 02)

The first step was to ensure the log format was consistent and could support ranking and exposure evaluation. After sanity checks (Notebook 01), I built an impression-level representation (Notebook 02) where each impression is linked to: (1) the user who received it, (2) the full candidate slate, and (3) which items were clicked. This step is crucial because all later analyses (exposure shares, candidate-vs-TopK comparisons, event studies,

and mitigation evaluation) require a consistent mapping **impression** $\rightarrow$ **candidates** and **impression** $\rightarrow$ **clicks**.

Table I summarizes the resulting scale (users, impressions, items, and candidate slate sizes).

B. User profiling and the engagement lens (Notebook 03)

Notebook 03 motivates the rest of the audit by establishing two baseline facts about the logs. First, **politics is scarce** in the candidate stream: political items account for only a small share of exposure (about 4.47%), while non-political items dominate (about 95.53%), as seen in Table II. Political clicks are also a small fraction of total clicks (about 3.83%), and the political CTR¹ is lower than the non-political CTR (about 0.0346 vs 0.0407). This confirms that ideology-level analysis must be interpreted under **supply limitations** (there is relatively little political content to begin with but there is still enough political data for simpler analyses).

Second, political interaction is **concentrated in a minority of users**. To make differential exposure measurable, I introduce an engagement lens: users with sustained political interaction are grouped as *engaged*, and all others as *other*. This cohort split is used throughout the following notebooks to quantify exposure gaps (engaged vs other) and to frame mitigation goals as balancing two risks: **saturation** (too much politics for engaged users) versus **deprivation** (too little politics exposure for other users when supply exists).

C. Baseline exposure and utility (Notebook 04)

Before diagnosing bias, it is necessary to define a baseline recommender and confirm that conclusions are not artifacts of a single scoring rule. In Notebook 04 I evaluated two simple baselines used throughout the project:

- **Popularity-based ranking** (a non-personal baseline),
- **Category-preference personalization** (a simple personalized baseline).

a. Political exposure imbalance across users. A key result from the baseline analysis is that political exposure in Top- K is not uniform across cohorts: politically engaged users receive substantially more political recommendations than other users (Figure 1). This motivates the first mitigation family later in the report: *reduce the engaged-other exposure gap without harming ranking utility*.

b. Why utility metrics matter here. Throughout the mitigation experiments, I use offline ranking utility metrics (MRR@ K and nDCG@ K) computed from dev clicks. Even when the goal is fairness/exposure, maintaining

similar MRR/nDCG provides evidence that the mitigation does not trivially destroy relevance.

D. Temporal stability and the limits of short-horizon reinforcement (Notebook 05)

After analyzing differences between politically engaged users and others at a higher level (focusing on overall political exposure and how different recommendation baselines behave without distinguishing ideologies) I next examine what happens *inside the dataset* at the impression level.

Specifically, I look at the candidate slates actually available to each user after a click (i.e., what the system could have recommended at the next step). The goal is to understand how the dataset’s own filtering and collection process shapes the available content over time, because any constraints in candidate availability directly affect:

- the Top- K exposure patterns observed earlier, and
- any later conclusions about reinforcement or ideology imbalance.

In practice, this is a short-horizon, timeline-style analysis: I follow groups of users across consecutive impressions and compare the candidate composition at time t and $t+1$, conditioned on what was clicked at t , to see whether the *availability of candidates* changes in a way that could explain or limit the exposure effects seen in ranked recommendations.

After establishing that politically engaged users tend to receive different levels of political exposure (Notebooks 03-04), Notebook 05 moves from aggregate averages to the *impression timeline*. I reconstruct per-user sequences of impressions and compare what happens immediately *after* different types of interactions, using a **binary** label (politics vs. non-politics)(Figure 2).

a. Main takeaway. The temporal analysis provides observational evidence consistent with **short-horizon reinforcement at the topic level**: impressions following political interactions tend to be associated with a higher political share for the next few steps compared to non-political baselines. This does not establish causality, but it motivates focusing subsequent analyses on **short horizons** (especially $t \rightarrow t+1$) and on mechanisms that can explain *why* exposure shifts (candidate supply vs. ranking).

b. Practical limitation that guides the rest of the project. Many users have short timelines and political events are unevenly distributed, which makes long-lag analyses (e.g., $t+2$, $t+3$, ...) noisier and less reliable. Therefore, later notebooks introduce ideology labels (Notebook 06) and test exposure mechanisms primarily at $t \rightarrow t+1$ and at the *candidate vs. Top- K* interface (Notebook 07).

E. Ideology labeling audit and supply limitations (Notebook 06)

In the previous notebooks, I analyzed in depth how political exposure differs between two user groups (politically engaged users vs. others). The next step is to go beyond the politics/non-politics split and examine the *composition of political content* itself: how different ideologies (left/center/right) are distributed in the dataset, and whether exposure to each ideology varies across user types and recommendation stages.

Notebook 06 was created to answer a key feasibility question:

Even if the ranking algorithm were perfectly fair, is there enough ideological variety in the *available supply* (candidate pool) to actually rank fairly?

To address this, Notebook 06 audits the ideology-labeling pipeline and quantifies: (i) **label coverage** (how many political items receive a usable ideology label), (ii) **class imbalance** (how skewed the labeled distribution is), and (iii) **label reliability** (how confident we can be in the labels).

a. Supply constraint (availability bottleneck). A recurring pattern in later analyses is that **center-labeled political content is structurally scarce**. This matters because fairness constraints at ranking time can only operate on what is actually present in the candidate set. In other words, if a user’s candidate slate contains very few (or zero) center items, then a constraint such as “always show at least one center item in Top- K ” is either: (i) **impossible** (no center items exist to select), or (ii) achievable only by forcing unrealistic behavior (e.g., repeatedly inserting low-relevance items or injecting external content). Therefore, any fairness or diversity intervention must be **conditional on availability** (i.e., enforce it only when the candidate pool contains the required items).

b. Why the main analysis focuses on center vs. non-center. Notebook 06 also highlighted a practical modeling trade-off: while a 3-class ideology scheme (left/center/right) is conceptually appealing, the **left vs. right separation was less reliable than desired**(Figure 3). To avoid building the main conclusions on noisier labels, I prioritized the **more robust distinction between center and non-center**, which showed stronger labeling consistency in the audit. As a result, the project’s core findings and mitigations are framed mainly around *center underrepresentation* and *non-center dominance*, where the labeling signal is more trustworthy.

The left/center/right analyses were still explored as an **additional, secondary investigation** to illustrate the broader ideological picture, but they are treated as less central due to labeling uncertainty and reduced confidence in fine-grained class comparisons.

F. Supply vs. ranking and short-horizon dynamics (Notebook 07)

Building on Notebook 06, I focus first on the most reliable ideological signal available in this dataset: whether political content is **center** or **non-center**.

a. Candidate slate vs. Top-K: isolating ranking effects. For the same impressions, I compute ideology composition among labeled political items (i) in the full candidate slate and (ii) in the ranked Top-K list. I report the shift as a percentage-point difference, which highlights *ranking effects* given the same supply.

b. Main result: center is suppressed beyond scarcity. Center is already rare in the candidate pool (Table III), but it becomes even rarer after ranking. Within the labeled political subset, center represents only a small fraction of candidates (roughly 5–6%), and its share drops further in Top-K (around 1–2%). This yields a large negative Δpp for center, indicating **ranking suppression on top of supply limitation**. Equivalently, non-center items occupy a larger share in Top-K than they had in candidates.

c. Feasibility matters (supply-aware interpretation). A key practical constraint is that center is not present in most candidate slates. In my runs, center existed in the candidate pool for only about 9–10% of impressions. Therefore, any center-focused mitigation must be **conditional on availability**: if a slate contains no center items, reranking alone cannot surface center without injecting new content.

d. Short-horizon transition analysis: adjacent impressions (t and $t+1$). To test whether user interactions are immediately followed by changes in what is available, I run a short-horizon, impression-adjacent analysis comparing the candidate slate at time t with the next impression $t+1$ for the same user. Impressions are grouped by what was clicked at t (e.g., center click vs. non-center political click vs. non-political click). The main conclusion is that there is **no clear evidence of immediate “center reinforcement” at the candidate stage** in $t+1$: even after center clicks, the next candidate slate does not reliably increase center availability. This result is also limited by scarcity: center-click events are very rare (Table IV), so estimates for that cohort are noisy.

e. Robustness across Top-K cutoffs. To ensure the suppression pattern is not an artifact of one cutoff, I repeat the candidate-vs-Top-K comparison for multiple values of K (e.g., $K \in \{5, 10, 20\}$). The qualitative effect persists: shorter Top-K lists disproportionately lose center relative to its already-low supply.

f. Left/center/right (secondary analysis). Available in the Appendix H.

G. Mitigations and trade-offs (Notebook 08)

The final stage evaluates mitigation strategies that are realistic under supply constraints and can be tested of-

fline.

1. Mitigation A: exposure-gap balancing (engaged vs other)

A first mitigation targets the **political exposure gap** between engaged and other users. The goal is not to eliminate politics for engaged users, but to reduce saturation while also avoiding deprivation for other users. I implemented a **soft floor/cap reranking** policy that:

- gently **caps** politics in Top-K for engaged users, and
- enforces a minimal **floor** of politics exposure for other users *when feasible* (i.e., when political items exist in the candidate slate).

Empirically, this reduced the engaged-other gap substantially (Table V) while preserving offline ranking utility (MRR/nDCG changes were negligible in my evaluation).

2. Mitigation B: “min-center when feasible” reranking

To address center underrepresentation, I tested a conditional constraint: **if center exists in the candidate slate, enforce at least one center item in Top-K**. This policy is explicitly supply-aware: it triggers only when center is available.

The mitigation greatly increases center presence in Top-K *in those impressions where center is feasible*, and it improves labeled coverage inside Top-K. Importantly, offline ranking metrics changed only slightly, suggesting the constraint can improve representational balance without a large relevance penalty.

a. Important interpretation detail. Because center is feasible in only a small fraction of impressions, some aggregated percentages can look surprising. The correct reading is: *when center is present, the policy forces center into Top-K, when center is absent, the Top-K is unchanged*. Therefore, the overall effect depends strongly on the feasibility rate.

Additional mitigations (supply bottleneck) and longer-horizon and intervention-style mitigations that were not tested are available in Appendix I.

H. Limitations and social impact

a. Key limitations. The most important limitations are: (i) **label incompleteness/noise** (mitigated by explicit coverage reporting and audits), (ii) **center scarcity** (limits event-study power and robustness of center-focused mitigation evaluation), and (iii) **offline evaluation constraints** (MRR/nDCG from clicks are an imperfect proxy for societal desirability).

b. Social impact and what the public should be aware of. News recommenders operate at the intersection of machine learning and democratic life: ranking policies influence what users repeatedly see and may shape political awareness over time. The results highlight two risks:

- **Exposure imbalance across users:** politically engaged users may be saturated with politics, while less engaged users may be under-exposed, reducing opportunities for civic information.
- **Under-representation of minority viewpoints:** if one ideology class is scarce or systematically ranked lower, Top- K feeds may omit that viewpoint, producing a skewed political spectrum.

These effects matter even if the system optimizes engagement, because engagement-driven ranking can amplify dominant content and narrow the range of viewpoints users encounter.

V. CONCLUSIONS AND FUTURE WORK

A. Conclusions

This project audited political exposure bias in a news recommendation setting, focusing on whether recommendation pipelines can amplify what users already interact with and create uneven exposure patterns. Using the MIND interaction logs, I reconstructed impression-level candidate slates and clicks, defined an engagement lens (politically engaged vs. other), and compared what is *available* (candidate slate) to what is *shown* (ranked Top- K).

a. Key takeaways.

- **Exposure is uneven across user types.** Even when overall political content is a small fraction of the feed, politically engaged users receive substantially higher political exposure than other users. This highlights a fairness risk of *saturation* for engaged users and *deprivation* for others.
- **Underrepresentation can be caused by both supply and ranking.** Ideology labeling and subsequent analysis show that some viewpoints (especially *center*) are already scarce in the candidate pool, and ranking can further suppress them in Top- K relative to their availability. This means that even a “cheap” reranker cannot fix what does not exist in the slate.
- **Short-horizon reinforcement signals are limited by scarcity.** Impression-adjacent analyses (t to $t+1$) are feasible at scale for broad categories (politics vs. non-politics), but ideology-specific event studies (e.g., center clicks) are highly underpowered because such events are rare. As a result, the strongest, most reliable conclusions

in this dataset come from *supply-vs-ranking* comparisons and group-level exposure gaps rather than multi-step causal claims.

- **Supply-aware post-processing mitigations can reduce measured gaps with minimal utility loss.** The tested reranking policies (soft floor/-cap for politics exposure across cohorts; “min-center when feasible” for center visibility) reduce exposure imbalance and improve representational presence *when the relevant items are available*, while producing only negligible changes in offline ranking metrics (MRR/nDCG).

B. Future Work

The results suggest several concrete directions that would strengthen both the evidence and the practical mitigation story:

- **Increase or diversify supply for robust center-focused evaluation.** Since center is scarce in the candidate pool, meaningful evaluation of center-aware policies requires either (i) more center content in the slate, (ii) additional labeled sources, or (iii) an experimental setup that injects/collects more center interactions.
- **Improve ideology labels and quantify uncertainty.** Future work should expand the manual audit, measure agreement more systematically, and treat ideology labels as uncertain (e.g., use confidence thresholds or calibration) so downstream analyses can report sensitivity to label noise.
- **Move from one-step effects to history-aware policies.** The most realistic interventions are longer-horizon: exposure balancing over windows, saturation control, and periodic diversity injections that respond to accumulated exposure rather than a single click. These strategies are especially important when rare events make $t+2$ and $t+3$ analyses unreliable.
- **Evaluate beyond offline utility proxies.** MRR/nDCG capture click-alignment, but societal desirability involves viewpoint diversity and informedness. Future work should include additional metrics (e.g., exposure diversity, concentration indices) and, ideally, online or user-study evaluation to test whether mitigations improve informational outcomes without harming user experience.
- **Formalize mitigation as multi-objective optimization.** A principled next step is to frame ranking as a trade-off between relevance and exposure constraints (e.g., fairness-of-exposure objectives), with constraints activated conditionally on availability.

Appendix A: Metrics and Definitions

1. CTR

¹ CTR (Click-Through Rate) = $\# \text{clicks} / \# \text{impressions}$.

2. Notebook 04

Utility is evaluated on dev impressions using clicks as binary relevance:

- **MRR@K**: rewards placing the first clicked item high in the ranking,
- **nDCG@K**: rewards ranking all clicked items high with logarithmic discount.

These metrics are later reused to check whether exposure-mitigation reranking produces only small utility changes.

Appendix B: More about Preprocessing and derived tables (Notebooks 01-02)

Notebook 01 (Sanity check) Some of the checks done were:

- missing values and invalid IDs,
- impression counts and candidate slate sizes,
- basic distribution checks to ensure the dev split is usable for offline evaluation.

Notebook 02 (Click table construction) standardizes the logs into analysis-ready tables:

- an impression-level table: $imp \rightarrow (user, candidates)$,
- a click table: $(imp, news_id, clicked)$,
- aggregated sets: $clicked(imp)$ used as relevance labels for MRR/nDCG.

Appendix C: Ideology labeling and label-quality audit (Notebook 06)

Coverage and Imbalance reported:

- **Label coverage**: the fraction of political items that receive a valid 3-class label;
- **Class distribution**: the relative frequency of left-/center/right within the labeled subset.

Appendix D: Supply vs ranking effect on ideology exposure (Notebook 07)

In this notebook we compare two distributions computed over the same impressions:

- **Candidate distribution**: ideology proportions among labeled political items in the full candidate slate.
- **Top- K distribution**: ideology proportions among labeled political items in the ranked Top- K list (for various K values).

I report shifts as **delta in percentage points**:

$$\Delta_{pp}(c) = \text{TopK}\%(c) - \text{Candidate}\%(c),$$

which isolates ranking effects given the same supply.

This notebook also includes a lightweight robustness check across $K \in \{5, 10, 20\}$ to verify whether suppression/amplification patterns persist as the visible list grows.

Appendix E: Short-horizon event study (Notebook 07)

I split events into three cohorts:

- **center_click**: at least one clicked item at t is political and labeled *center*,
- **noncenter_click**: at least one clicked item at t is political and labeled *non-center* (and no center was clicked),
- **nonpolitical_click**: clicked items at t are non-political.

For each cohort, I compute candidate-slate shares at t and at $t + 1$:

- **politics_share**, **center_share**, **noncenter_share**, **nonpolitics_share**,
- and **center_within_politics** (center fraction among political candidates only).

Important limitation. Center-click events are rare (very small n), so conclusions about center reinforcement are interpreted cautiously and framed as “no strong evidence of short-term reinforcement” rather than definitive causal effects.

Appendix F: Mitigations (Notebook 08)

What was done:

- **exposure-gap mitigation**: soft cap/floor reranking to shrink the engaged-vs-other political exposure gap when feasible,

- **center-visibility mitigation:** conditional reranking to ensure at least one center item in Top- K when center exists in candidates,
- evaluate utility cost using MRR@ K and nDCG@ K , plus diagnostics (how often the constraint is feasible and how often Top- K changes).

Appendix G: Results

Appendix H: Additional Ideology labeling audit (Notebook07)

a. Left/center/right (secondary analysis). As an additional (lower-priority) view, I also analyze the 3-class distribution (left/center/right) where available. The qualitative takeaway aligns with the binary results: Top- K becomes **more right-skewed** than the candidate pool, and **center becomes even more underrepresented**. Because Notebook 06 showed lower confidence for fine-grained left vs. right separation, I treat these 3-class findings as **supporting evidence**, while anchoring the main conclusions on the more reliable **center vs. non-center** signal.

Appendix I: Additional mitigations (Notebook 08)

1. Additional mitigations not tested (supply bottleneck)

Notebook 07 highlighted that center-focused mitigations are fundamentally constrained by **supply**. Beyond the tested strategies, several practical options exist:

- **Exploration boost for rare classes:** slightly up-weight center items when present (e.g., multiplicative boost or additive prior).
- **Within-politics diversity constraint:** ensure at least m center items in Top- K *when available*.
- **Two-stage ranking:** rank politics/non-politics first, then re-rank politics with a center/non-center balance objective.
- **Data mitigation:** improve label coverage and collect/inject more center content to make evaluation meaningful.

I did not test these quantitatively because center items are too scarce in the candidate pool for a robust offline comparison. A meaningful evaluation would require higher center availability or an experimental setup that collects more center interactions.

2. Longer-horizon and intervention-style mitigations

Most analyses in this project focus on short-horizon effects ($t \rightarrow t+1$) and candidate-to-Top- K shifts. However, exposure imbalance can also be addressed with **history-aware policies** that respond to exposure patterns over time:

- **Exposure balancing over windows:** track exposure over the last N impressions and gently increase scarce types when available.
- **Saturation control:** apply soft penalties when one topic/ideology dominates a user’s recent feed.
- **Periodic diversity injections:** occasionally recommend content that is intentionally more diverse than the user’s usual feed.
- **Adaptive constraints:** floors/caps triggered by exposure history and availability rather than always-on rules.

These strategies treat bias as an accumulated exposure dynamic rather than a single-click immediate consequence, which is especially relevant when rare events limit short-horizon inference.

Appendix J: Tables and Plots

a. Notebook 01-02

TABLE I: Dataset summary statistics (DEV impressions).

Metric	Value
Rows (candidates shown)	5,843,444
Clicks	236,344
CTR	0.040446
Unique users	50,000
Unique items	20,288
Impressions	156,965
Avg candidates / impression	37.227688
Median candidates / impression	24.0

b. Notebook 03

TABLE II: Overall exposure and engagement for politics vs non-politics.

Category	Exposure share	Click share	CTR
Non-politics	0.9553	0.9617	0.0407
Politics	0.0447	0.0383	0.0346

c. Notebook 04

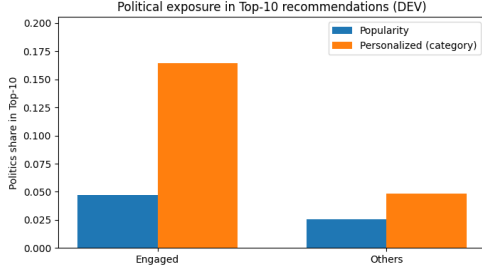


FIG. 1: Political exposure in Top-10 recommendations on the DEV split, comparing two baselines (Popularity vs Personalized-by-category) across engagement cohorts. Engaged users receive a substantially higher share of political items than other users under both baselines, indicating an exposure gap that later analyses decompose into candidate-supply vs ranking effects.

d. Notebook 05

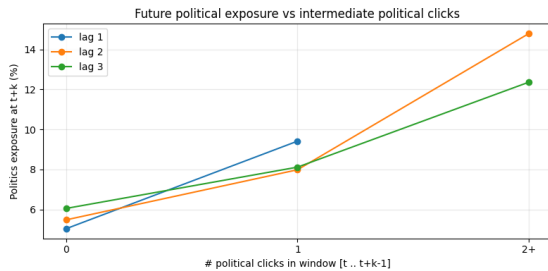


FIG. 2: Short-horizon reinforcement signal: future political exposure at time $t+k$ increases with the number of political clicks in the preceding window $[t, t+k-1]$. Lines correspond to different horizons (lag 1–3). Across lags, users with 2+ intermediate political clicks receive markedly higher subsequent political exposure, consistent with temporal reinforcement of political consumption.

e. Notebook 06

Confusion matrix (rows=true, cols=pred):

	pred left	pred_center	pred_right
true_left	10	3	6
true_center	2	11	2
true_right	3	1	7

Classification report:

	precision	recall	f1-score	support
left	0.67	0.53	0.59	19
center	0.73	0.73	0.73	15
right	0.47	0.64	0.54	11
accuracy			0.62	45
macro avg	0.62	0.63	0.62	45
weighted avg	0.64	0.62	0.62	45

FIG. 3: Manual audit of the ideology classifier on $n=45$ labeled news items. Confusion matrix (rows=true, cols=pred) and per-class precision/recall/F1 show moderate agreement (accuracy ≈ 0.62), with most confusion occurring between left/right and neighboring classes. This motivates treating 3-way ideology results as noisier and prioritizing more robust analyses (e.g., center vs. non-center and supply-vs-ranking comparisons).

f. Notebook 07

TABLE III: Within-politics ideology distribution used in the audit. Center-labeled political items are scarce ($\approx 7.7\%$) relative to non-center ($\approx 92.3\%$), which constrains the feasibility of center exposure in Top- K and motivates supply-vs-ranking analysis.

Ideology (within politics)	Proportion	Percent (%)
Non-center	0.9233	92.33
Center	0.0767	7.67

g. Notebook 08

TABLE V: Exposure-gap mitigation example (politics share in Top- K). The policy reduces the engaged-other gap by decreasing engaged exposure and increasing other exposure when feasible.

Group	Baseline politics share	Mitigated politics share
Engaged	0.1568	0.1227
Other	0.0445	0.0699

TABLE IV: Total political exposures appearing in users’ Top- K recommendation lists, aggregated over impressions, by cohort and baseline. Cohorts involving center clicks are small relative to the overall population, highlighting event rarity and limited power for center-specific temporal analyses.

Cohort	Popularity (count)	Popularity (%)	Preference (count)	Preference (%)
Engaged (no center clicks)	23	0.53	96	1.14
Engaged (with center clicks)	12	0.28	24	0.28
Other	4322	99.20	8336	98.58
Total	4357	100.00	8456	100.00

Note: Counts reflect the total number of political items appearing in Top- K lists aggregated over impressions (not the number of users).

¹ K. Järvelin and J. Kekäläinen, *ACM Transactions on Information Systems* **20**, 422 (2002).

² A. Singh and T. Joachims, in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (2018).

³ A. J. B. Chaney, B. M. Stewart, and B. E. Engelhardt, in

Proceedings of the 12th ACM Conference on Recommender Systems (RecSys) (2018).

⁴ Microsoft, “Mind: Microsoft news dataset,” <https://msnews.github.io/> (2020), accessed: 2026-01-02.

⁵ Bucket Research, “[bucketresearch/politicalBiasBERT](https://bucketresearch.com/politicalBiasBERT/),” Hugging Face model card (2023), accessed: 2026-01-02. Label order in model card: [0]=left, [1]=center, [2]=right.